

Practical # 09

Objective:

Implement linear search algorithm and analyze its performance for different cases.

Theory:

Linear search algorithm finds a given element in a list of elements with $O(n)$ time complexity where n is total number of elements in the list. This search process starts comparing search element with the first element in the list. If both are matched then result is element found otherwise search element is compared with the next element in the list. Repeat the same until search element is compared with the last element in the list, if that last element also doesn't match, then the result is "Element not found in the list". That means, the search element is compared with element by element in the list.

Linear search is implemented using following steps...

Step 1 - Read the search element from the user.

Step 2 - Compare the search element with the first element in the list.

Step 3 - If both are matched, then display "Given element is found!!!" and terminate the function

Step 4 - If both are not matched, then compare search element with the next element in the list.

Step 5 - Repeat steps 3 and 4 until search element is compared with last element in the list.

Step 6 - If last element in the list also doesn't match, then display "Element is not found!!!" and terminate the function.

In this lab, students will learn how to:

- **Develop** C++ programs to implement the Linear Search algorithm.
- **Apply** Linear Search to arrays, strings, and structures.
- **Analyze** the performance of Linear Search in terms of best, average, and worst cases.
- **Evaluate** the number of comparisons required for different input sizes.

C++ program: Write C++ program for linear search algorithm.

```
// C++ Program to Implement Linear Search
#include<iostream>
using namespace std;
int main()
{
    int arr[10], i, num, index;
    cout<<"Enter 10 Numbers: ";
    for(i=0; i<10; i++)
        cin>>arr[i];
    cout<<"\nEnter a Number to Search: ";
    cin>>num;
    for(i=0; i<10; i++)
    {
        if(arr[i]==num)
        {
            index = i;
            break;
        }
    }
    cout<<"\nFound at Index No."<<index;
    cout<<endl;
    return 0;
}
```

OUTPUT

```
Enter 10 Numbers:
10
11
12
1
3
6
5
22
7
8
Enter a Number to Search: 1
Found at Index No.3
```

Review Questions/ Exercise:

1. Write a C++ program to implement Linear Search for an array of integers.
2. Modify the Linear Search program to search for a character in a string and display its index if found.
3. Develop a program to implement Linear Search in an array of structures (e.g., student records with roll number and name).
4. Analyze the performance of Linear Search for best case, average case, and worst case inputs. Represent results in tabular form.
5. Construct a program to count the number of comparisons made during Linear Search and compare results for small (10 elements), medium (100 elements), and large (1000 elements) input sizes.
6. Develop a program that allows user to enter multiple keys and searches them sequentially in an array using Linear Search.

Name: _____

Roll #: _____

Date: _____

Subject Teacher

